NNWA Nutrition and Wellness Academy
पोषण और कल्याण अकादमी

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Protein, *decoded.*



How much you need, where to find it in everyday Indian food, and how to spread it across your day.

A plain-language guide

From NNWA (Nutrition & Wellness Academy). General information for healthy adults, not medical advice.

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How to use this guide

Read it once, then use the workbook in chapter 07 for a week. Every fact comes from a government source, listed in the bibliography on page 29.

Before you start

General information for healthy adults, not medical advice. If you are pregnant, breastfeeding, under 18, older, unwell or training heavily, speak to a qualified healthcare professional.

01

What protein *does.*

Before the numbers, here is what protein does in the body and why food has to supply part of it every day.

01 What protein does
02 How much you need
03 Where to find it
04 Spread it across your day
05 Myths, checked
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01 | Six jobs, every day.

Protein is made of building blocks called amino acids. The body uses 20 of them. It can make 11 itself; the other nine, called essential (indispensable) amino acids, must come from food.

9

essential amino acids we must get from food

01 Builds and repairs

Protein is needed for growth and to replace body tissues as they are broken down.

02 Runs reactions

The body uses protein to make enzymes, which drive digestion and metabolism.

03 Sends signals

Protein is also needed to make hormones, the body's chemical messengers.

04 Builds cells

Proteins are core building parts of every living cell, including cell membranes.

05 Carries

Haemoglobin, a carrier protein in the blood, takes oxygen to the tissues.

06 Gives energy

About 4 kcal per gram. Enough carbohydrate and fat lets protein be used for building.

Why daily matters

Body tissues are constantly broken down and replaced, so meals have to keep supplying protein. About half the body's protein is in muscle; the rest is in bone, cartilage and skin.

01 | Enough grams, not always enough quality.

ICMR-NIN's guidelines note that an Indian adult's daily diet often already provides 60 to 70 g of protein, more than most people need. The real challenge is quality: getting all nine essential amino acids, which depends on the mix of foods.

10–15%

of daily energy should come from protein

Why muscle matters at every age

01 **Stronger bones**
Keeping good muscle mass helps keep bones strong.

02 **Easier joints**
Good muscle helps joints move freely, with less strain on knees and hips.

03 **Better blood sugar**
More muscle improves insulin sensitivity and lowers the risk of diabetes, stroke and heart attack.

04 **Use it or lose it**
Without physical activity, protein is not used for muscle, and muscle wastes away.

ICMR-NIN's own example

For a person weighing 65 kg, the guidelines put the average requirement at 43 g of protein a day and the RDA at 54 g, whatever their gender or activity level.

02

How much do you *need?*

One number covers most healthy adults. This chapter gives you that number, a table to find yours, and the situations where it changes.

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02 | One number to remember.

0.83 g per kg of body weight, a day

This is the Recommended Dietary Allowance (RDA) for healthy Indian adults set by ICMR-NIN. It covers 97.5% of healthy people, so most need a little less. On a mostly cereal diet with little pulse, milk or egg, ICMR-NIN puts it at 1 g per kg.

QUICK MATHS	Weight	× 0.83	= grams	60 → 50
	in kg	the RDA	per day	kg to g, example

When the number changes

Pregnancy and breastfeeding	Needs go up. ICMR-NIN adds extra protein in the 2nd and 3rd trimesters and while breastfeeding.
Children and teenagers	Needs are set by age and growth stage, not by the adult figure.
Older adults	ICMR-NIN advises foods rich in good-quality protein, plus regular activity to keep muscle.
Kidney disease	High-protein diets need care if your kidneys are affected. Follow your doctor's advice.
Heavy training	The adult RDA does not rise with activity; most athletes meet it from food. See chapter 05.

General information only, not medical advice.

02 | Your weight, your grams.

37 g	**at 45 kg**	a day
42 g	**at 50 kg**	a day
46 g	**at 55 kg**	a day
50 g	**at 60 kg**	a day
58 g	**at 70 kg**	a day
66 g	**at 80 kg**	a day

Daily protein at the 0.83 g per kg RDA, rounded to the nearest gram.

02 | ICMR-NIN's numbers, by life stage.

Adults		
Man 65 kg reference weight		**54 g**
Woman 55 kg reference weight		**46 g**
Pregnancy and breastfeeding		
2nd trimester + 9.5 g		**55 g**
3rd trimester + 22 g		**68 g**
Feeding, 0–6 months + 16.9 g		**63 g**
Feeding, 6–12 months + 13.2 g		**59 g**
Children and teenagers		
Child, 4–6 years		**16 g**
Child, 7–9 years		**23 g**
Girl, 16–18 years		**46 g**
Boy, 16–18 years		**55 g**

RDA in grams of protein a day, from ICMR-NIN's 2020 requirements table, rounded to the nearest gram. Pregnancy and breastfeeding add to the 55 kg woman's 45.7 g. For your own needs in these stages, follow your doctor.

03

Where to *find* it.

Protein is already in Indian kitchens: in dal, curd, paneer, eggs, soya, nuts, fish and chicken. Here is roughly how much each serving brings, and how to pair foods well.

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03 | Roughly how much, per serving.

Plant foods

Soybean 25 g dry	9 g
Cooked dal 1 katori, from 30 g raw	7 g
Peanuts 30 g	7 g
Chana dal 30 g raw	6 g
Cooked rice 1 cup, from 50 g raw	4 g
Roti 1 medium, 30 g atta	3 g

Dairy

Paneer 50 g	9 g
Milk 200 ml	7 g
Curd 1 katori, 150 g	5 g

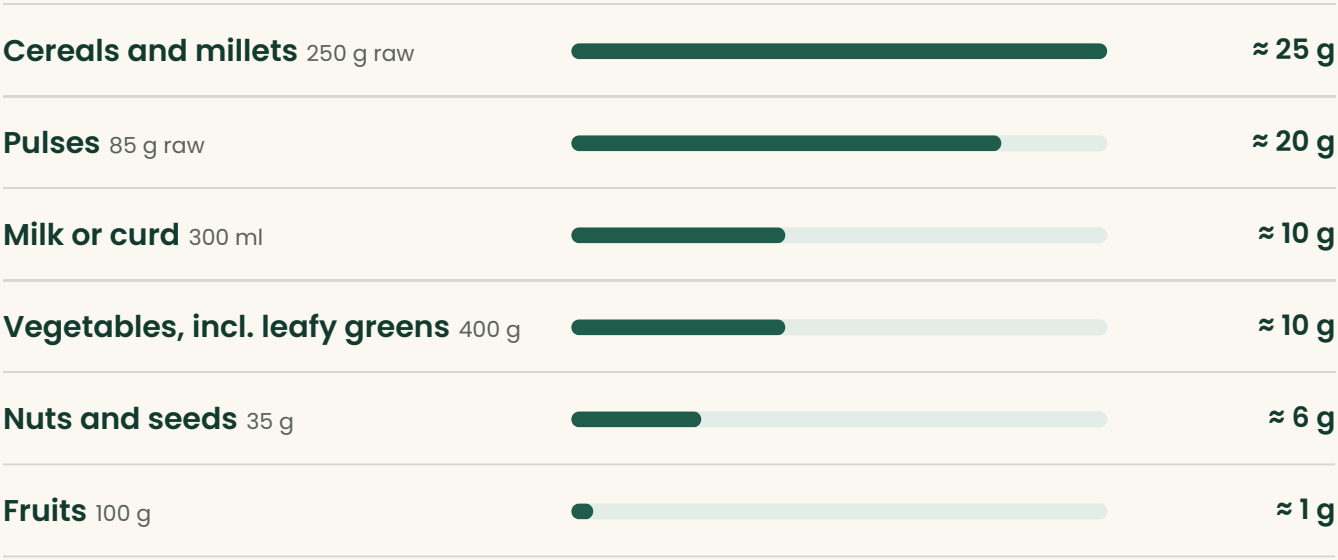
Eggs, fish and meat

Chicken breast 100 g raw	22 g
Fish (rohu) 100 g raw	20 g
Egg 1 egg, about 50 g	7 g

Approximate, rounded values calculated by NNWA from per-100 g figures in ICMR-NIN's Indian Food Composition Tables; curd from the milk and curd figures in ICMR-NIN's 2024 guidelines. Actual amounts vary with variety, recipe and portion. Bars scale to 9 g; chicken and fish run past the scale.

03 | India's plate for the day.

ICMR-NIN's "My Plate for the Day" builds a 2,000 kcal vegetarian day from everyday Indian food groups. Together they give about 72 g of protein, 15% of the day's energy.



The non-vegetarian plate

Swap to 55 g of pulses plus 70 g of chicken, meat or eggs, and the day gives about 79 g. ICMR-NIN suggests 700 to 900 g of fish, poultry or lean meat a week.

The Indian kitchen helps

Fermenting (idli, dosa, dhokla) and sprouting, both everyday Indian methods, improve how well nutrients are digested and absorbed.

03 | Better together.

Cereals such as rice and wheat are low in the essential amino acid lysine. Pulses are low in methionine. Eaten in the same meal, each fills the other's gap, which is why so many Indian staples already pair them.

01 **Dal and chawal**
Rice with any dal.

02 **Rajma or chole with rice**
Kidney beans or chickpeas.

03 **Idli or dosa with sambar**
Rice and urad batter, toor dal sambar.

04 **Khichdi**
Rice and moong dal in one pot.

05 **Roti with chana or dal**
Wheat with pulses.

06 **Millet roti with dal**
Bajra, jowar or ragi with pulses.

Complete on their own

Milk, curd, paneer, eggs, fish and meat provide good-quality protein with all nine essential amino acids. Adding a little of any of them to a cereal meal also lifts its protein quality. Soya is often counted here too.

03 | Dal and chawal



PROTEIN INFORMATION	≈ 11 g	1 katori	1 cup	Paired
	protein	cooked dal	cooked rice	pulse + cereal

On the plate

- Dal, from 30 g raw: ≈ 7 g
- Rice, from 50 g raw: ≈ 4 g
- Add a katori of curd: ≈ 5 g

Good to know

Rice is low in lysine and dal is low in methionine. Eaten together, each fills the other's gap, so the meal's protein is better used than either food alone.

Try: rajma-chawal, chole with rice, or khichdi.

Approximate values calculated by NNWA from IFCT 2017 per-100 g figures; actual amounts vary by dal, recipe and portion. Photo via Pexels.

03 | Paneer, portioned



PROTEIN INFORMATION	≈ 9 g	50 g	Dairy	Quality
	protein	paneer	food group	good-quality protein

On the plate

Paneer, 50 g: ≈ 9 g

With 2 rotis: + ≈ 6 g

Shown here in a gravy

Good to know

Paneer is a good-quality dairy protein and an easy vegetarian option. It is also rich in fat, so a measured 50 g portion keeps a meal balanced. Grilled, bhurji or in palak works better than heavy, creamy gravies every day.

Try: paneer bhurji with roti, paneer tikka, or palak paneer.

Approximate values calculated by NNWA from IFCT 2017 per-100 g figures; actual amounts vary by milk and recipe. Photo via Pexels.

03 | Eggs, two ways



PROTEIN INFORMATION	≈ 7 g	1 egg	Eggs	Quality
	per egg	about 50 g	food group	good-quality protein

On the plate

1 egg, 50 g: ≈ 7 g

2 eggs at breakfast: ≈ 13 g

With 1 roti: + ≈ 3 g

Good to know

An egg is one of the simplest sources of good-quality protein. The white and the yolk both carry protein, but almost all the fat is in the yolk. Eating the whole egg gives you both.

Try: egg curry with roti, egg bhurji with vegetables, or a boiled egg.

Approximate values calculated by NNWA from IFCT 2017 per-100 g figures. Photo via Pexels.

04

Spread it *across* the day.

A protein food at every meal is easier than one big serving, and it fits how most Indian households already eat.

01 What protein does
02 How much you need
03 Where to find it
04 **Spread it across your day**
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04 | Little and often.

One illustrative day for a 60 kg adult, whose RDA works out to about 50 g. This day lands a little above it. It is an example to adapt, not a meal plan.

BREAKFAST	**A glass of milk with a boiled egg** Swap in curd, besan chilla or sprouts.	14 g
LUNCH	**Two rotis, a katori of dal, a katori of curd, sabzi** Cereal and pulse in the same meal.	18 g
SNACK	**A small handful of peanuts or sprouts chaat** Easy to carry; watch the portion.	7 g
DINNER	**Rice with a katori of rajma and a katori of curd** Or 50 g paneer with 2 rotis; fish or chicken works too.	15 g

DAY TOTAL, APPROXIMATELY

54 g



Values are approximate, from the serving table on page 11 (rajma from IFCT 2017).

04 | Three Indian homes, four days.

Illustrative examples, not real people. Each "needs" figure is the ICMR-NIN RDA for that person; each total uses the serving values on page 11. Every day here is met with ordinary home food and no supplement.

MEERA, 55 KG Vegetarian, desk job, Pune	Curd with 2 rotis; dal, rice and curd at lunch; peanuts at 5 pm; paneer with 2 rotis at dinner. Needs 46 g	49 g
ARJUN, 65 KG Eats fish, Kolkata	2 eggs and a roti; 2 cups of rice with dal and a piece of rohu; dal with 2 rotis at night. Needs 54 g	64 g
FATIMA, 55 KG 7 months pregnant, Hyderabad	Milk, an egg and 2 rotis; dal, rice, curd and 2 rotis; peanuts and milk; paneer, dal and 2 rotis. Needs 68 g	78 g
RAJU, 16 Plays school cricket, Nagpur	Milk and an egg; 2 cups of rice, dal and curd; peanuts after practice; 3 rotis with rajma. Needs 55 g	56 g

Arjun's 64 g sits in the 60 to 70 g ICMR-NIN says many Indian adults already eat. In pregnancy, ICMR-NIN also advises daily iron and folic acid tablets; Fatima follows her doctor. Values are approximate NNWA calculations.

05

Myths, *checked.*

Ten things people often repeat about protein, set against what ICMR-NIN's guidance for Indians says.

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05 | What the evidence says.

01

~~More protein always means more muscle~~

Training builds muscle. Protein supports it, up to a point.

Muscle grows with resistance exercise plus enough protein and energy. Research cited in ICMR-NIN's 2024 guidelines finds intakes above about 1.6 g/kg a day add no further gains.

02

~~Vegetarians cannot get enough protein~~

A varied vegetarian diet can meet the RDA.

Cereals and pulses together, with milk, nuts and seeds, can supply all the essential amino acids.

03

~~Plant protein does not count~~

It counts. It works best in combination.

Single plant foods may be low in one amino acid. Pairing cereals and pulses fills the gap (page 13).

04

~~You need a shake after every workout~~

For most people, regular meals do the job.

ICMR-NIN advises against protein supplements for building muscle (chapter 06).

05

~~Protein is only for gym-goers~~

Everyone needs it, every day.

The RDA applies to every healthy adult, whatever their activity level; see page 04 for its roles.

Check a claim yourself

Ask who is saying it, whether a named source such as ICMR-NIN backs it, and whether it promises a quick result. Sound nutrition advice rarely does.

05 | Myths heard in Indian homes.

06

~~Dal alone gives complete protein~~

Dal works best with a cereal.

Pulses are low in methionine. ICMR-NIN suggests cereals and pulses in about a 3:1 ratio by raw weight, as in dal-chawal or khichdi.

07

~~Indians do not eat enough protein~~

Most get the grams; quality is the gap.

ICMR-NIN notes an adult diet often provides 60 to 70 g a day. The challenge is getting all nine essential amino acids from the right mix.

08

~~Whey is harmless because it comes from milk~~

Most people do not need it at all.

Whey is rich in branched-chain amino acids (BCAAs). ICMR-NIN notes recent evidence that BCAAs may raise the risk of some lifestyle diseases.

09

~~Protein damages healthy kidneys~~

The caution is for existing kidney disease.

ICMR-NIN says high protein intake may worsen kidney function in people who already have kidney disease, so they need medical guidance.

10

~~The more meat, the better~~

There is a sensible weekly amount.

ICMR-NIN suggests 700 to 900 g of fish, poultry or lean meat a week, alongside cereals, pulses and vegetables.

06

Supplements, *carefully.*

Protein powders are everywhere. Here is what India's dietary guidelines say about them, and what to check if you are still considering one.

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06 | Start with food.

What the guidelines say

ICMR-NIN's Dietary Guidelines for Indians (2024) advise against protein supplements for building muscle and recommend meeting needs through food combinations. They link large amounts taken as a sole supplement over long periods to bone mineral loss, and warn high protein can worsen existing kidney disease.

If you are still considering one

01

Speak to a professional first

Especially if you have any medical condition, are pregnant, or are under 18. This is NNWA's general advice.

02

Check the FSSAI licence number

Packaged foods sold in India must show the FSSAI logo and licence number on the label. Buy from a traceable seller.

03

Read the label, not the claims

Labels must list ingredients and nutrition facts, including protein, per 100 g or per pack. Check added sugar too.

04

Be wary of big promises

FSSAI rules require claims to be truthful and scientifically backed, and food generally may not claim to treat or cure disease. Treat big promises as a warning sign.

General information only, not medical advice.

06 | Reading an Indian label.

FSSAI's rules decide what a protein product's pack must show and what it is allowed to claim. Five things to look for.

01

Green dot or brown triangle

A green filled circle in a green square marks vegetarian food; a brown filled triangle in a brown square marks non-vegetarian food.

02

“Source of protein” has a meaning

A solid food needs at least 10% of the RDA per 100 g to say “source of protein”, and 20% to say “high” or “rich in” protein.

03

Per 100 g, then % per serve

Nutrition facts appear per 100 g, 100 ml or pack, with each serve's share of daily values based on a 2,000 kcal diet.

04

No “doctor recommended”

A label may not say or imply that the food is recommended or approved by doctors, nutritionists or health professionals.

05

Food first is the law's view too

Claims may not suggest that a balanced and varied diet cannot give the body the nutrients it needs.

General information only, not medical or legal advice.

07

Your *workbook.*

Two pages to fill in. Track a week of protein habits, then keep the checklist where you will see it.

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07 | A week of habits.

Tick a box each time the habit happens. The ticks in the first column show one example day.

HABIT	M	T	W	T	F	S	S
Protein at breakfast milk, curd, egg, chana	✓						
Protein at lunch dal, rajma, paneer, fish	✓						
Protein at dinner one serving, not the day's worth	✓						
Cereal + pulse together dal-chawal, khichdi, idli-sambar	✓						
Dairy or soya serving curd, milk, paneer, tofu							
Nuts or seeds as a snack a small handful	✓						
Label check before supplements FSSAI number, serving, sugar	✓						

Easiest meal this week

One change for next week

07 | Six habits to keep.

- [ ] **I know my number** — ONCE
  Worked out below, and checked with a professional if page 07 applies to me.
- [ ] **Protein at every meal** — DAILY
  Breakfast, lunch and dinner each include a protein food.
- [ ] **A cereal and pulse together** — DAILY
  Dal-chawal, khichdi, idli-sambar or roti-chana.
- [ ] **A dairy or soya serving** — DAILY
  Curd, milk, paneer, tofu or soya chunks.
- [ ] **Nuts or seeds as a snack** — MOST DAYS
  A small handful, not the whole packet.
- [ ] **Label check before any supplement** — EVERY TIME
  FSSAI number, serving size, added sugar, professional advice.

My weight _____ **kg × 0.83 =** _____ *grams a day*

Bibliography.

Every fact, figure and recommendation in this guide is drawn from the government publications below. Serving values are NNWA's rounded calculations from sources 2 and 3.

1 **ICMR-NIN. A Brief Note on Nutrient Requirements for Indians, the Recommended Dietary Allowances (RDA) and the Estimated Average Requirements (EAR). ICMR-National Institute of Nutrition, 2020. Protein: Table 2a and its footnote.**
Used on pages 07, 08, 09, 18, 19, 21.

2 **ICMR-NIN. Dietary Guidelines for Indians. ICMR-National Institute of Nutrition, 2024. Protein: pp. 6–8, 14–15, 56, 57–61 (Guideline 8), 81 and 100–101.**
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3 **ICMR-NIN. Indian Food Composition Tables (IFCT). ICMR-National Institute of Nutrition, 2017. Codes A015, A019, B001, B020, B021, B024, H012, L002, L003, M001–M003, N003, S006.**
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5 **FSSAI. Food Safety and Standards (Advertising and Claims) Regulations, 2018. Food Safety and Standards Authority of India. Regulations 4 and 10 and Schedule I.**
Used on pages 24, 25.

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Important

This guide is general educational information for healthy adults. It is not medical advice and does not diagnose, treat or prevent any condition. If you are pregnant or breastfeeding, under 18, an older adult, training heavily, or living with kidney, liver or any other medical condition, consult a qualified healthcare professional before changing your diet. NNWA (Nutrition & Wellness Academy) is an education provider.

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KEEP LEARNING

Understand the science on your *plate.*

NNWA (Nutrition & Wellness Academy) is a 100% online academy for nutrition, wellness and fitness certification. This guide is one of a series of short, sourced nutrition basics.

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